DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. **Branch: Chemical/Petrochemical** **Semester: VI**

Subject Code & Name: BTCHC605 Plant Utilities & Plant Safety

Max Marks: 60 **Date: 26/08/22** **Duration: 3.45 Hrs.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Mar
Q. 1	**Solve Any Two of the following.**		
A)	Name the various water softening process. Explain ion exchange process with neat sketch.	**CO1**	**6**
B)	Write down the difference between Temporary and Permanent hardness. Explain in Brief scale and sludge formation in boiler	**CO1**	**6**
C)	What are the uses of air? explain briefly on compressed air, blower air and fan air	**C01**	**6**
Q.2	**Solve Any Two of the following.**		
A)	Briefly, explain different boilers accessories	**CO2**	**6**
B)	Describe in details Indian boiler act, 1923.	**CO2**	**6**
C)	What is steam generator and what is its function? What are uses of steam?	**CO2**	**6**
Q. 3	**Solve Any Two of the following.**		
A)	Write note on primary and secondary refrigerant and cryogenic refrigerator	**CO3**	**6**
B)	Write down short notes on refrigerant. A) refrigerant 500 B) Sulphur dioxide c) methylene chloride D) brine	**CO3**	**6**
C)	What are refrigerants? Explain criterion for selection of a good refrigerant.	**CO3**	**6**
Q.4	**Solve Any Two of the following.**		
A)	Write the hazards and remedies of acryl amide and anhydrous ammonia.	**CO4**	**6**
B)	Write short notes on- (a) butyl amine (b)acryl amide (d) allyl alcohol	**CO4**	**6**
C)	What are the safety parameters in process design? Explain safety parameters in the production of aromatic nitriles. And write down the examples of hazardous processes.	**C04**	**6**
Q. 5	**Solve Any Two of the following.**		
A)	Suggest a suitable control scheme for the safe operation of chemical reactor.	**CO1**	**6**
B)	Discuss the various methods available for the analysis of loss prevention in chemical plants.	**CO2**	**6**
C)	Give the types of fires, explain briefly four most common types of fire extinguishers.	**CO1**	**6**

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